Sanyam Mehta

sanyam@umich.edu · sanyam-mehta.github.io/about-me · LinkedIn · GitHub · Scholar

Research interests: world models, latent-action learning and planning, video generation, 3D robot perception.

Education

University of Michigan, Ann Arbor, M.S. in Robotics, GPA 4.0/4.0 — 2024 – 2026

Punjab Engineering College, B.Tech. in Computer Science & Engineering — 2016 – 2020

Exchange semester at IIIT-Delhi — 01/2020 – 05/2020

Research

Point Prompting: Counterfactual Tracking with Video Diffusion Models — 08/2024 – 01/2026

Sanyam Mehta*, Ayush Shrivastava*, Daniel Geng, Andrew Owens (*equal contribution) — *ICLR 2026*

University of Michigan, advised by Andrew Owens [paper] [project] [ICLR]

- Asked whether video diffusion models already know how points move: showed that off-the-shelf ones can track points with no training. The method draws a colored marker on a point in the first frame, regenerates the video, and follows the marker, comparing against an unedited generation to pick the marker out reliably.
- Beat the prior zero-shot state of the art by **42%** in Average Jaccard and 19% in occlusion accuracy, comparable to trained self-supervised trackers. It works across different video diffusion architectures.

JEVA: Joint Encoders for Variational Action World Models [project]

Aditya Singh, **Sanyam Mehta**, Yixin Wang, Nima Fazeli, Bernadette Bucher — *Under review, 2026*

- Robot action labels are scarce but video is plentiful; JEVA learns a latent-action world model from mostly action-free video, training inverse dynamics and the world model together. With **5% of episodes labeled**, its policy matches or beats every baseline trained on 20%.
- Push-T often has several distinct ways to reach a goal. Tested whether a discrete codebook of latent actions would keep the planner's averages from landing in regions the model has no representation for, especially with little data. Continuous latents still worked better.
- Helped compare MPPI with CEM for planning in the latent space; CEM won consistently, which raised the question of how well-behaved the latent action space really is.
- Set up a Franka arm and collected real-robot trajectories; the method didn't hold up on the real setup, so the submission was scoped to simulation.

Independent & Ongoing Research

Planning in JEPA-style latent-action world models, *independent project* — 10/2026 – present

- Asked whether CEM's 23-point lead over MPPI on LeWM / Push-T meant the latent action space was irregular. Traced most of it to CEM shrinking its sampling noise; with the same adaptive noise, MPPI goes **66% to 84%**.
- Separated the two planners' differences with a 2×2 experiment that swapped how each weights its samples and whether its noise shrinks, then tested it with runs whose predictions were written down beforehand.

Robot-agnostic navigation from video models, *advised by Bernadette Bucher* — 07/2025 – 01/2026

- Started with a diffusion policy that predicts navigation waypoints from the instruction and a 192×192×768 OpenSeg feature map of the scene, cropped around the robot's known pose and fed in through cross-attention. Trained on about 10.8K R2R-VLN-CE episodes (Matterport3D in Habitat) on up to 8 L40S GPUs.
- The policy only saw the present: it predicted waypoints from the current view, with nothing about where the robot had already been, though following an instruction depends on that history. Its map input was also very large to condition on.
- That led to video models, which condition on past frames and text by construction. Given the history, the view facing the goal and the instruction, a video model generates the future, and matching frames recovers the camera motion. Navigation becomes pose estimation that any robot can follow with IK.
- Built the first stage: collected about 10.7K Habitat episodes and fine-tuned SigLIP to pick, out of 24 views, the one leading toward the goal. On unseen scenes its pick is within 15° of the right direction **76%** of the time and within 30° 84% of the time, vs 12.5% and 21% by chance.

Industry Experience

Gather AI, Pittsburgh, PA 2021 – 2024, 2026 – present

A deployed fleet of autonomous drones that scan warehouse racks for inventory, and newer forklift-mounted cameras.

Machine Learning Scientist *(rejoined after M.S.)* 07/2026 – present

- Built a 3D method to count boxes in a pallet's partial top layer, which production had been counting as zero. It lifts box masks into 3D with VGGT, measures each box's height, and clusters the heights into layers. Kept it geometric so any failure can be traced to one step and fixed.
- On a 500-pallet holdout of single-partial-layer stacks (the case the method targets), exact case-count accuracy went from **0% to 58%**. Across 8 production missions covering all pallet types, bins sent to human review dropped from 438 to 358.
- Ported it to production with a teammate in August 2026, after checking that the port reproduced the research counts on all 30 real bins tested.
- Added per-message tracing to the forklift's ROS2 vision pipeline, so the team can see which node drops a barcode detection and replay any day of logs offline.

ML Engineer*, mentored by Dr. Daniel Maturana* 09/2022 – 08/2024

- Trained a U-Net segmentation model and combined it with monocular depth estimates to infer 3D occupancy from 2D warehouse images, reaching 95%+ accuracy.
- Trained a Panoptic-DeepLab model to refine ROI boundaries, which cut the error rate of pallet product identification from drone images by 90%.
- Compared ORB-SLAM and DROID-SLAM on off-the-shelf drones and found DROID-SLAM still holds up at 5 FPS. Proposed a hybrid navigation approach that could save about $40K in instrumentation per facility.

ML-Ops Engineer 04/2021 – 09/2022

- Built the data engine and model deployment system for the drone fleet. It processed 2.5M+ pallet locations and made inventory counting 15x faster.
- Trained and deployed a network that finds empty rack locations from 2D images at 97%+ precision on 1.2M+ samples. Its deployment led to a promotion to ML Engineer.
- Ran a data-annotation process with teams in the US and Kenya, writing the labeling instructions and steadily improving the quality of the labels.

Teaching & Service

Graduate Student Instructor, ROB 550 Robotic Systems Lab, University of Michigan 01/2025 – 05/2026

- Taught robot-arm programming from scratch and particle-filter SLAM with LiDAR and odometry. Coached student teams on design choices like state machines and multiprocessing.

Reviewer: ICLR 2027, CoRL 2026, CoRL 2026 TAG Workshop

Skills

Languages: Python, C++, C

ML / vision: PyTorch, diffusion policies, video diffusion models, SigLIP, OpenSeg, VGGT, Depth Anything, U-Net, Panoptic-DeepLab, CEM and MPPI planning

Robotics / infra: ROS2, MCAP, Habitat, ORB-SLAM, DROID-SLAM, particle-filter SLAM, Docker, Singularity, MLflow, Weights & Biases

Training: multi-GPU PyTorch (DDP) on SLURM clusters